

AI NAVAL MONITORING

A PROJECT REPORT

Submitted by

LAKSHNAASREE S V (8115U24AM032)

in partial fulfilment for the award of the course

AMB 1221-MACHINE LEARNING FUNDAMENTALS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

MAY 2026

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Under the Guidance of

Mrs. C.MUTHUSELVI

Department of Artificial Intelligence and Data Science

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BONAFIDE CERTIFICATE

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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ANNA UNIVERSITY, CHENNAI



DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

- M1** To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.
- M2** To foster Experiential learning equips students with engineering skills to tackle real-world problems.
- M3** To promote collaborative innovation in AI, machine learning, and related research and development with industries.
- M4** To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- PEO1** Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.
- PEO2** Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.
- PEO3** Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- PSO1** Expertise in tailoring ML algorithms and models to excel in designated applications and fields.
- PSO2** Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The main objective of the Naval Monitoring System is to ensure real-time monitoring of critical parameters such as voltage, current, temperature, and pressure to maintain system efficiency and safety. The system aims to detect faults like electrical leakage, overheating, and abnormal conditions at an early stage to prevent damage and accidents. It also focuses on analyzing wave energy using camera-based technology for better environmental understanding. Additionally, the system provides data visualization through interactive graphs, helping users make informed decisions. An automated alert mechanism is included to notify users when any parameter exceeds safe limits. The system maintains proper data logging with timestamps for future reference and analysis. It also ensures secure access through user authentication using login credentials. Finally, the system enables report generation by allowing users to download data in CSV format, including system status and suggested solutions.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
The main objective of the Naval Monitoring System is to ensure real-time monitoring of critical parameters such as voltage, current, temperature, and pressure to maintain system efficiency and safety. The system aims to detect faults like electrical leakage, overheating, and abnormal conditions at an early stage to prevent damage and accidents. It also focuses on analyzing wave energy using camera-based technology for better environmental understanding. Additionally, the system provides data visualization through interactive graphs, helping users make informed decisions. An automated alert mechanism is included to notify users when any parameter exceeds safe limits. The system maintains proper data logging with timestamps for future reference and analysis.	PO1-3 PO2-3 PO3-3 PO4-3 PO5-3 PO6-3 PO7-3 PO8-3 PO9-3 PO10-3 PO11-3	PSO1-3 PSO1-3

Note: 1- Low, 2- Medium, 3- High

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LIST OF ABBREVIATIONS

S. NO.	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	IOT	Internet of Things
3	UI	User Interface
4	CSV	Comma Separated Values
5	ML	Machine Learning
6	GUI	Graphical User Interface
7	API	Application Programming Interface
8	V	Voltage
9	T	Temperature
10	P	Pressure
11	RT	Real Time

CHAPTER 1

INTRODUCTION

1.1 Objective

The objective of this project is to design and develop an advanced and intelligent Naval Monitoring System that provides continuous real-time monitoring, analysis, and control of critical parameters in marine environments. In modern naval operations, systems are exposed to complex challenges such as unpredictable environmental conditions, equipment failures, undetected leakages, and lack of efficient and centralized monitoring solutions. This project aims to address these issues by integrating sensor-based data acquisition, real-time processing, and AI-driven analytical techniques to deliver accurate, reliable, and timely insights into system performance. The system continuously monitors essential parameters such as voltage, current, temperature, and pressure to ensure operational safety, stability, and efficiency.

In addition to basic monitoring, the system is designed to detect abnormalities such as sudden fluctuations, overheating, excessive pressure, and electrical leakage at an early stage, thereby preventing potential hazards and reducing maintenance costs. A key feature of the system is the integration of camera-based technology, which enables real-time analysis of sea wave patterns to estimate wave energy and identify dangerous environmental conditions. This enhances situational awareness and supports safer naval operations. The system also provides advanced data visualization through interactive and dynamic graphs, allowing users to easily interpret trends, patterns, and variations in system behavior. The project also emphasizes the development of a user-friendly and interactive interface, making the system accessible even to users with minimal technical expertise. Overall, the objective is to combine modern technologies such as IoT, artificial intelligence, and web-based visualization to create a smart, scalable, efficient, and reliable naval monitoring solution that enhances safety, improves decision-making, and optimizes overall system performance.

1.2 Overview

The Naval Monitoring System is a web-based application developed using Python and the Dash framework, integrating real-time data processing, sensor monitoring, and visualization techniques to provide a comprehensive marine system analysis platform. The system is structured into multiple modules including user authentication, real-time parameter monitoring, graphical visualization, wave energy detection, and data logging, ensuring a smooth and organized workflow. Users can securely log into the system using authorized credentials and access a dashboard that displays key parameters such as voltage, current, temperature, and pressure in real time. Based on these inputs, the system continuously analyzes the data to detect abnormal conditions such as electrical leakage, overheating, or system faults.

Additionally, the system incorporates camera-based functionality to capture and analyze sea wave patterns, enabling estimation of wave energy and identification of potential hazards. The graphical interface presents the monitored data through interactive charts, helping users easily understand system behavior and trends. Furthermore, the system maintains a digital log that records timestamp, system status, and suggested solutions, which can be downloaded in CSV format for further analysis. This combination of monitoring, analysis, visualization, and reporting makes the system highly efficient, intelligent, and user-friendly, providing a modern solution for naval system management.

1.3 Purpose and Importance

The purpose of the Naval Monitoring System is to develop an intelligent and efficient solution for real-time monitoring of critical parameters such as voltage, temperature, current, and pressure, ensuring system safety and reliability. The project aims to detect abnormal conditions at an early stage and provide immediate voice alerts to enable quick response and prevent potential failures. Its importance lies in reducing human operational

efficiency. Additionally, the system supports data logging and CSV download, which helps in analysis, reporting, and maintenance planning. Overall, the project plays a significant role in enhancing monitoring accuracy and demonstrates including timestamps, parameter values, and system status. This logged data acts as an important source for analysis, performance evaluation, and maintenance planning. Users can download this data in CSV the practical application of modern technologies in real-world naval and industrial environments.

1.4 Data Source Description

The Naval Monitoring System utilizes multiple data sources to ensure accurate and real-time monitoring of system parameters. The primary data source consists of sensor-based inputs, which include important parameters such as voltage, temperature, current, and pressure. These values are either collected from physical sensors connected to the system or generated through simulated data for testing and development purposes. The data is continuously updated at regular intervals to reflect real-time conditions and ensure effective monitoring. In addition to numerical data, the system also generates internal data during operation, such as system status (normal or abnormal), error conditions, and alert messages. This data is essential for identifying faults and ensuring system safety. These values are either collected from physical sensors connected to the system or generated through simulated data for testing and development purposes. The system processes all incoming data and compares it with predefined threshold values to detect any abnormal behavior.

Furthermore, the system maintains a structured log of all collected data, including timestamps, parameter values, and system status. This logged data acts as an important source for analysis, performance evaluation, and maintenance planning. Users can download this data in CSV format for further processing using external tools such as spreadsheets. Overall, the combination of sensor data, system-generated information, and stored logs provides a reliable and comprehensive data source for the Naval Monitoring System, enabling accurate monitoring, efficient fault detection, and informed decision-making.

1.5 Project Summarization

The Naval Monitoring System is an intelligent and efficient solution developed to enhance the safety and performance of marine systems through real-time monitoring and analysis. The project focuses on continuously tracking critical parameters such as voltage, temperature, current, and pressure to ensure proper system functioning and to prevent potential failures. By integrating sensor-based data collection with real-time processing, the system provides accurate and up-to-date information about system conditions. The project is designed with multiple modules, including user authentication, parameter monitoring, error detection with voice alerts, data logging, and CSV download functionality. These modules work together to create a seamless and user-friendly system that allows users to monitor system performance effectively. The voice alert mechanism ensures that users are immediately notified of any abnormal conditions, improving response time and safety. Additionally, the system provides interactive visualization through graphs and dashboards, making it easier for users to understand trends and variations in the monitored parameters.

The data logging feature enables users to store and analyze historical data, which is useful for performance evaluation and maintenance planning. The ability to export data in CSV format further enhances the system's usability and flexibility. Overall, the project successfully combines real-time monitoring, intelligent alerting, data management, and user-friendly design into a single platform. It reduces manual effort, minimizes errors, and improves decision-making, making it a reliable and practical solution for naval and industrial monitoring applications. It also improves decision-making by presenting real-time data in a clear and understandable format, helping users take quick and effective actions during critical situations. The Naval Monitoring System is an intelligent and efficient solution developed to enhance the safety and performance of marine systems through real-time monitoring and analysis. The project focuses on continuously tracking critical parameters such as voltage, temperature, current, and pressure to ensure proper system functioning and prevent potential failures.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction and Traditional Methods

Naval monitoring has been an essential aspect of marine operations for ensuring safety, efficiency, and proper functioning of equipment. In earlier days, traditional monitoring methods were primarily based on manual observation and standalone instruments such as analog meters and gauges to measure parameters like temperature, pressure, voltage, and current. These systems required constant human supervision and were highly dependent on the operator's experience and attentiveness. For example, issues such as overheating, or pressure imbalance could go unnoticed until they caused serious damage. Communication between different monitoring units was limited, and there was no centralized system to integrate and analyze data from multiple sources. As a result, decision-making was slow and inefficient. Although these methods were simple and cost-effective, they were not suitable for handling the increasing complexity and dynamic nature of modern naval systems.

2.2 Sensor Based and Intelligent Monitoring Techniques

With the advancement of electronics and computing technologies, modern naval monitoring systems have evolved to use sensor-based data acquisition and intelligent processing techniques. Sensors play a crucial role in continuously collecting real-time data pressure, and other environmental conditions. These sensors are connected to microcontrollers or computing systems that process the data, the integration of Internet of Things (IoT) technologies allows multiple sensors and devices to communicate with each other, creating a connected monitoring environment. Intelligent techniques, including basic artificial intelligence and machine learning algorithms, are used to analyze the collected data and identify patterns or anomalies. These systems can detect unusual behavior such as sudden voltage drops, abnormal temperature rise, or leakage conditions and generate alerts automatically.

2.3 Web-Based Systems and Visualization Techniques

The development of web technologies has significantly improved the way monitoring systems are designed and accessed. Web-based naval monitoring systems allow users to access real-time data through browsers without the need for specialized software. Frameworks such as Python-based Dash enable the creation of interactive dashboards that display data in the form of graphs, charts, and indicators. Visualization techniques play a key role in understanding complex datasets. Real-time line graphs, bar charts, and dynamic plots help users easily interpret variations in parameters like voltage, current, temperature, and pressure. These visual tools make it easier to identify trends, detect anomalies, and make informed decisions quickly. Web-based systems also support remote monitoring, enabling users to access the system from any location. They often include features such as user authentication, data logging, report generation, and cloud storage. These capabilities make the system more flexible, scalable, and user- friendly, enhancing its practical usability in real-world scenarios

2.4 Limitations of Existing Systems and Summary

Despite significant advancements in monitoring technologies, existing systems still face several challenges and limitations. Many traditional systems lack real-time processing and integration, while some modern systems focus only on specific aspects such as sensor monitoring or visualization without combining all functionalities into a single platform. Some systems require expensive hardware and complex installation procedures, making them less accessible for widespread use. In certain cases, systems may not provide accurate predictions or fail to detect complex anomalies due to limited analytical capabilities. Additionally, many systems do not include environmental monitoring features such as wave analysis, which is crucial for naval applications. Another major limitation is the lack of user-friendly interfaces in some existing systems, which makes them difficult to operate for non-technical users. Data management and reporting features are also limited in certain systems, reducing their effectiveness in long-term analysis and decision-making.

CHAPTER 3

PROJECT METHODOLOGY

This chapter describes the methodology used to design and develop the Naval Monitoring System. It explains the overall workflow, system architecture, and the integration of various modules required for real-time monitoring and analysis of marine parameters. The system is developed using a web-based approach with Python and the Dash framework. It also includes modules such as user authentication, real-time parameter monitoring, graphical visualization, wave energy analysis, alert generation, and data logging. This chapter provides a clear understanding of how the system operates and how each component works together to deliver a complete, intelligent, and user-friendly naval monitoring solution.

3.1 Proposed Workflow

The proposed workflow of the Naval Monitoring System defines the step-by-step process through which the system operates to ensure efficient monitoring, analysis, and reporting of marine parameters. The workflow begins with the user accessing the system through a secure login interface, where authentication is performed using valid credentials to ensure authorized access. Once the user successfully logs in, they are directed to the main dashboard, which serves as the central interface for monitoring and interaction. The next stage involves real-time data acquisition, where sensor inputs such as voltage, current, temperature, and pressure are continuously collected. In the absence of physical sensors, simulated data is generated to mimic real-world conditions. This data is then passed to the processing unit, where it is analyzed using predefined threshold values and basic intelligent techniques to detect any abnormal conditions such as overheating, excessive pressure, or electrical leakage. Overall, the proposed workflow integrates authentication, data collection, analysis, visualization, alert generation, and reporting into a seamless process, making the Naval Monitoring System efficient, reliable, and user-friendly.

3.2 Architectural Overview

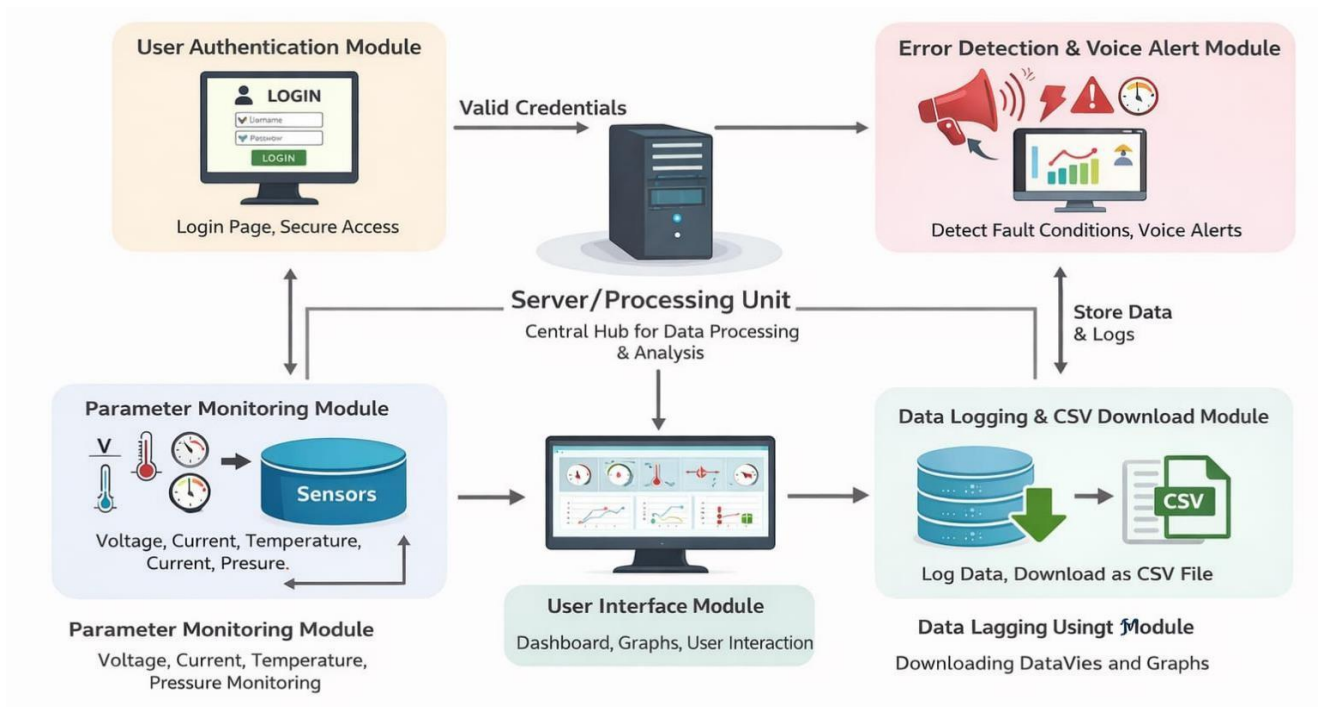


Fig 3.2.1 Architecture Diagram

The architecture diagram of the Naval Monitoring System represents the overall structure and flow of the system. It illustrates how different modules such as user authentication, parameter monitoring, voice alert system, and data logging are interconnected. The diagram shows how data is collected from sensors, processed in the system, and displayed on the user interface through real-time graphs. It also highlights how errors are detected and alerts are generated to notify users. Additionally, the architecture includes the CSV download module for storing and exporting data. Overall, the diagram provides a clear understanding of how all components work together to ensure efficient and reliable system operation.

3.3 Hardware and Software Requirements

The Naval Monitoring System is designed to operate on commonly available hardware with moderate computational requirements, ensuring smooth and efficient performance. A computer or laptop with at least an Intel i3 processor or equivalent and a minimum of 4GB RAM is sufficient to run the application, although higher specifications can enhance real-time data processing and visualization performance. The system may also require hardware components such as voltage, current, temperature, and pressure sensors for real-time data collection, along with a camera module to capture sea wave patterns for wave energy analysis. A stable internet connection and power supply are necessary for continuous monitoring and system accessibility. OpenCV or similar libraries are used for camera-based image processing and wave detection. The system stores data in CSV format for logging and reporting, and optionally, a database can be integrated for advanced storage needs. Development tools such as Visual Studio Code or PyCharm are used for coding, testing, and debugging the application. This combination of hardware and software ensures that the system is efficient, scalable, and user-friendly, making it suitable for real-time naval monitoring applications.

3.4 System Integration and Working

The Naval Monitoring System is developed by integrating multiple modules into a unified platform to ensure seamless operation and efficient real-time monitoring. The system combines data acquisition, processing, analysis, visualization, and reporting components, all connected through a web-based interface. The working of the system begins with user authentication, where authorized users log in using valid credentials to access the application. Once logged in, the user is directed to the main dashboard, which acts as the central control panel. The system then starts collecting data from various sources, including sensors and the camera module. Sensor data such as voltage, current, temperature, and pressure is continuously acquired and sent to the processing unit, while the camera captures real-time visuals of sea wave conditions

CHAPTER 4

RELEVANCE OF THE PROJECT

4.1 Why the Model Was Chosen

The Naval Monitoring System is designed using a combination of sensor-based monitoring, real-time data processing, and basic AI-driven analysis techniques. This approach was chosen because naval environments require continuous and accurate monitoring of multiple parameters such as voltage, current, temperature, and pressure. Instead of relying on complex and computationally expensive models, the system uses threshold-based analysis and simple intelligent techniques, which are efficient, fast, and reliable for real-time applications. The chosen model ensures immediate detection of abnormal conditions such as leakage, overheating, and pressure fluctuations without delay. Additionally, the integration of camera-based wave energy analysis enhances the system's capability by combining both numerical and visual data. This hybrid approach is suitable for real-time naval systems where speed, accuracy, and reliability are more important than computational complexity.

4.2 Comparison with Other Models

Compared to traditional monitoring systems, which rely on manual observation and standalone instruments, the proposed system offers automated, real-time monitoring with improved accuracy and efficiency. Traditional systems lack centralized control and are prone to human errors, whereas the proposed system integrates all functionalities into a single platform. When compared to advanced machine learning models, such as deep learning techniques, those models require large datasets, high computational power, and complex training processes. In contrast, the proposed system uses lightweight analysis techniques that do not require extensive training data, making it more practical and cost-effective for real-time implementation.. Additionally, some existing systems focus only on sensor data without considering environmental factors.

Feature	Threshold-Based System	Machine Learning Model	Deep Learning Model
Accuracy	Medium	High	Very High
Speed	Very Fast	Fast	Slow
Complexity	Low	Medium	High
Data Need	Very Low	Medium	High
Implementation	Easy	Moderate	Difficult
Real-Time Use	Excellent	Good	Limited
Cost	Low	Medium	High

4.3 Advantages and Disadvantages

Advantages:

- Provides real-time monitoring of multiple parameters.
- Enables early fault detection and improves safety.
- Combines sensor data and visual analysis for better accuracy.
- User-friendly interface with interactive graphical visualization.
- Supports data logging and report generation in CSV format.
- Cost-effective and easy to implement compared to complex systems.
- Reduces human effort and minimizes errors.

Disadvantages:

- Accuracy depends on the quality and calibration of sensors.
- Camera-based wave analysis may be affected by lighting and weather conditions.
- Limited use of advanced machine learning techniques for deeper predictions.
- Requires continuous power supply and system maintenance

4.4 Summary of Relevance

The Naval Monitoring System holds significant relevance in today's technologically advancing marine and naval environments, where safety, efficiency, and real-time decision-making are of utmost importance. Traditional monitoring methods are no longer sufficient to handle the increasing complexity of naval operations, as they lack automation, real-time analysis, and integrated system management. This project addresses these challenges by providing a modern, intelligent, and unified solution that enhances monitoring capabilities and operational reliability. The system is highly relevant due to its ability to continuously monitor critical parameters such as voltage, current, temperature, and pressure in real time. By ensuring constant observation of these parameters, the system helps in maintaining optimal performance and prevents unexpected failures. One of the key contributions of this project is its capability for early fault detection, which reduces the risk of equipment damage, system downtime, and potential hazards. This is particularly important in naval environments, where even minor faults can lead to serious consequences. Another important aspect of the system's relevance is the integration of both numerical and visual monitoring techniques. The inclusion of camera-based wave energy analysis provides additional environmental awareness, which is not commonly available in many existing systems. This feature enhances the system's ability to detect hazardous conditions such as strong waves or unusual patterns, thereby improving overall safety and situational awareness. The project is also relevant in terms of technological implementation, as it combines concepts from IoT, artificial intelligence, and web-based development into a single platform. This integration demonstrates the practical application of modern technologies in solving real-world problems. The use of interactive dashboards and graphical visualization makes the system user-friendly and accessible, allowing users to easily interpret complex data and make informed decisions quickly.

CHAPTER 5

MODULE DESCRIPTION

5.1 User Authentication Module

The User Authentication Module is a fundamental component of the system, responsible for ensuring secure and authorized access. Since the Naval Monitoring System deals with critical operational data, it is essential to restrict access only to permitted users. This module provides a login interface where users must enter valid credentials such as username and password. If incorrect credentials are entered, the system denies access and may prompt the user to re-enter valid details. This prevents unauthorized users from accessing sensitive monitoring information. In addition to security, this module enhances system reliability by maintaining controlled access. It can also be extended to support advanced features such as encrypted passwords, multi-user login systems, role-based access (admin/user), and login attempt tracking. The authentication process is designed to be simple, efficient, and user-friendly, ensuring that users can quickly access the system without unnecessary complexity while still maintaining a high level of security.

5.2 Parameter Monitoring Module (Voltage, Temperature, Current, Pressure)

The Parameter Monitoring Module is the core and most important module of the Naval Monitoring System. It is responsible for continuously collecting, updating, and displaying key operational parameters such as voltage, temperature, current, and pressure. These parameters are essential indicators of the system's health and performance. The module receives input either from real-time sensors or simulated data sources. The data is updated at regular intervals to reflect real-time conditions. Continuous monitoring helps in identifying fluctuations, trends, and abnormal variations in system behavior. For example, a sudden rise in temperature may indicate overheating, while irregular current values may suggest leakage or circuit faults.

This module ensures that all parameter values are accurately captured and transmitted to the processing unit without delay. It also prepares the data for visualization and further analysis. The real-time nature of this module makes it highly valuable, as it allows users to monitor the system continuously without interruption. Moreover, this module forms the foundation for other modules such as error detection and alert systems. Without accurate monitoring, it would not be possible to identify faults or ensure system safety. Hence, this module plays a critical role in maintaining system stability, performance, and reliability.

5.3 Error Detection and Voice Alert Module

The Error Detection and Voice Alert Module is designed to enhance system safety by identifying abnormal conditions and immediately notifying the user. This module continuously receives data from the monitoring module and compares the parameter values with predefined threshold limits. These thresholds represent safe operating conditions for each parameter. When a parameter exceeds its safe range—for example, high temperature, abnormal voltage, excessive pressure, or unusual current flow—the system detects it as an error or fault condition. Once an error is identified, the module triggers a voice alert that audibly informs the user about the specific issue. This feature is highly beneficial in real-time environments where users may not constantly monitor the screen. The voice alert system improves response time by instantly drawing attention to critical issues. It reduces dependency on visual monitoring and ensures that problems are not overlooked. Additionally, this module can be expanded to include different types of alerts such as alarms, pop-up notifications, or automated system shutdowns in severe conditions. Overall, this module plays a crucial role in preventing system failures, reducing risks, and ensuring operational safety. It acts as an intelligent warning system that helps users take timely corrective actions.

5.4 Data Logging and CSV Download Module

The Data Logging and CSV Download Module is responsible for recording and storing all system data for future analysis and reference. Every parameter value, along with its corresponding timestamp, system status, and detected errors, is stored in a structured format. This creates a detailed record of system activity over time. This module is essential for maintaining historical data, which can be used for performance evaluation, fault analysis, and maintenance planning. By analyzing past data, users can identify patterns, trends, and recurring issues, which helps in improving system efficiency and reliability. In addition to data storage, the module provides a CSV download feature. Users can export the recorded data into a CSV file, which can be easily opened using spreadsheet software such as Excel. This allows for further analysis, visualization, and reporting outside the system. The logging system ensures data consistency, accuracy, and accessibility. It also supports debugging and system improvement by providing detailed insights into system behavior. Overall, this module enhances the analytical capability of the system and supports informed decision-making.

5.5 User Interface and Visualization Module

The User Interface and Visualization Module is responsible for providing an interactive and user-friendly environment for system interaction. It acts as the front-end of the application, where users can view system data, monitor performance, and interact with different features. This module includes the dashboard, graphical displays, and control elements. It presents real-time data using dynamic graphs, charts, and visual indicators. Visualization techniques such as line graphs help users easily understand trends and variations in parameters like voltage, temperature, current, and pressure. A well-designed interface reduces complexity and improves usability. Even users with minimal technical knowledge can operate the system efficiently. The layout is designed to be clear, organized, and responsive, ensuring smooth navigation and better user experience.

Additionally, the interface integrates all modules, allowing seamless interaction between data monitoring, alert systems, and data logging features. It acts as a bridge between the user and the system, ensuring that all information is presented in an understandable and accessible manner.

5.6 Machine Learning Module

The Machine Learning Module is an advanced component of the Naval Monitoring System designed to enhance the system's intelligence by enabling predictive analysis and smart decision-making. While the existing system monitors real-time parameters and detects errors based on predefined thresholds, this module adds the capability to learn from historical data and identify patterns that may not be easily detected using traditional methods. This module utilizes historical data collected from the system, including parameters such as voltage, current, temperature, and pressure, along with system status (normal or abnormal). The data is first pre processed, where missing values are handled, noise is reduced, and the data is structured in a suitable format for model training. Feature selection is performed to identify the most relevant parameters that influence system behaviour. A suitable machine learning algorithm, such as Decision Tree, Random Forest, or Logistic Regression, can be used to train the model. The trained model learns the relationship between input parameters and system conditions, enabling it to classify whether the system is operating normally or approaching a fault condition. In more advanced cases, regression models can be used to predict future parameter values based on current trends. Once trained, the model is integrated into the system to perform real-time predictions. As new data is received from the monitoring module, the machine learning model analysis it and potential issues before they occur. For example, it can identify gradual increases in temperature or pressure that may lead to system failure in the near future. This predictive capability allows users to take preventive actions rather than reacting to faults after they occur.

CHAPTER 6

RESULT AND DISCUSSION

6.1 Performance Analysis

The performance of the Naval Monitoring System is evaluated based on its ability to provide accurate, real-time monitoring, efficient fault detection, and user-friendly interaction. The system demonstrates strong performance in continuously tracking critical parameters such as voltage, current, temperature, and pressure. The real-time data updating mechanism ensures that users receive instant information about system conditions without delay, which is essential for maintaining safety in naval environments. One of the key strengths of the system is its fast response time in detecting abnormal conditions. By comparing parameter values with predefined threshold limits, the system is able to quickly identify faults such as overheating, abnormal voltage fluctuations, excessive pressure, and current leakage. The integration of the voice alert module further enhances performance by immediately notifying users about errors, reducing the chances of unnoticed failures. This feature significantly improves reaction time and helps in preventing potential damage. The system also performs effectively in terms of data visualization. The use of interactive graphs and charts allows users to easily understand system behaviour and trends over time.

This improves decision-making and enhances the overall usability of the system. The dashboard is responsive and updates dynamically, ensuring a smooth user experience. Another important aspect of performance is data management. The system successfully records all monitored data along with timestamps, system status, and detected issues. The CSV download feature allows users to export this data for further analysis and reporting. This capability is useful for maintaining historical records, conducting performance evaluations, and identifying long-term trends. In terms of reliability, the system performs consistently under continuous operation.

6.2 User Feedback

The Naval Monitoring System was evaluated based on user interaction and feedback to understand its usability, effectiveness, and overall performance in real-time monitoring scenarios. Users who tested the system included individuals with both technical and non-technical backgrounds, which helped in assessing the system from different perspectives. Overall, the feedback received was positive, particularly regarding the system's simplicity and ease of use. Users appreciated the user-friendly interface, which allowed them to navigate through the dashboard and understand the displayed data without difficulty. The graphical representation of parameters such as voltage, current, temperature, and pressure was found to be highly effective, as it simplified complex data and made it easier to identify trends and variations over time. One of the most appreciated features of the system was the real-time monitoring capability.

Users found it highly useful that the system continuously updated parameter values, providing instant information about system conditions. This feature was especially important in ensuring safety and preventing unexpected failures. Additionally, the voice alert mechanism received positive feedback, as it enabled users to quickly recognize abnormal conditions without constantly monitoring the data. The data logging and CSV download feature was also well-received by users. They found it beneficial for maintaining records, past data, and preparing reports. The ability to export data in a simple format made it convenient for further analysis using external tools such as spreadsheets. This feature added value to the system by supporting both real-time monitoring and long-term data storage. However, users also provided some suggestions for improvement. A few users mentioned that the system could be enhanced by incorporating more advanced predictive features using machine learning for better fault prediction. Others suggested improving the visual design of the dashboard to make it more attractive and informative. Some users also recommended adding mobile compatibility and remote access features to increase flexibility and usability.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Outcomes

The Naval Monitoring System project has been successfully designed and developed to provide an efficient, reliable, and intelligent solution for real-time monitoring of critical marine parameters. The primary objective of the system was to ensure continuous monitoring of parameters such as voltage, current, temperature, and pressure, and to detect abnormal conditions at an early stage. The outcomes of the project clearly demonstrate that these objectives have been effectively achieved. The system successfully implements a real-time monitoring mechanism that continuously updates parameter values, allowing users to track system performance without delay. This ensures better operational awareness and helps in maintaining system stability. One of the key outcomes is the accurate detection of abnormal conditions using predefined threshold values.

The system is capable of identifying issues such as overheating, voltage fluctuations, excessive pressure, and current leakage, thereby preventing potential system failures. Another significant outcome of the project is the integration of the voice alert mechanism, which enhances system responsiveness. This feature ensures that users are immediately notified about critical issues, even if they are not actively monitoring the dashboard. As a result, the system improves safety and reduces the chances of unnoticed errors. The project also achieves effective data management through its logging and CSV download functionality. All monitored data, along with timestamps and system status, is recorded and can be exported for further analysis. This outcome supports long-term performance evaluation, troubleshooting, and documentation. The ability to analyze historical data adds significant value to the system. The development of a user-friendly interface is another important outcome. The interactive dashboard with graphical visualization allows users to easily understand system behavior and trends.

7.2 Future Scope and Enhancement

The Naval Monitoring System developed in this project provides a strong foundation for real-time monitoring and analysis of marine parameters. However, there are several opportunities for future improvements and enhancements that can further increase the efficiency, intelligence, and applicability of the system in real-world naval environments. One of the major areas for future enhancement is the integration of advanced machine learning and deep learning techniques.

By incorporating predictive models, the system can analysis historical data to forecast potential failures before they occur. This will enable predictive maintenance, reduce downtime, and improve overall system reliability. The system can also be enhanced to automatically learn from new data and improve its accuracy over time.

Another important enhancement is the integration of IoT (Internet of Things) technology. By connecting real-time sensors directly to the system through IoT devices, the monitoring process can become more accurate and fully automated. This would eliminate the need for simulated data and allow the system to operate in real-world environments with live sensor inputs. Wireless communication between sensors and the monitoring system can further improve flexibility and scalability.

The system can also be extended by improving the camera-based module for advanced environmental analysis. Techniques such as image processing and computer vision can be used to accurately analysis wave patterns, detect obstacles, and identify hazardous conditions in the marine environment. This would enhance situational awareness and contribute to safer naval operations.

In terms of user interface, the system can be enhanced by developing a more interactive and visually appealing dashboard with additional features such as customizable views, advanced analytics, and real-time notifications. Mobile application support can also be introduced, allowing users to monitor the system remotely from smartphones or tablets, thereby increasing accessibility and convenience.

Another possible enhancement is the integration of cloud computing for data storage and processing. Storing data in the cloud would allow secure access from multiple locations and enable large-scale data analysis. It would also improve data backup and recovery capabilities. Additionally, implementing role-based access control and stronger security mechanisms can further enhance system security.

The system can also be expanded to include automated control mechanisms. For example, in case of critical conditions, the system can automatically shut down certain operations or activate safety measures without human intervention. This would further reduce risks and improve system efficiency.

The Naval Monitoring System can be further enhanced by integrating advanced technologies such as machine learning and deep learning to enable predictive analysis and early fault prediction.

The system can be improved by incorporating IoT-based real-time sensors for accurate and automated data collection in real-world environments. Additional features such as mobile application support and remote monitoring can increase accessibility and user convenience.

The camera module can be enhanced using image processing techniques for better wave analysis and hazard detection. Cloud integration can also be implemented for secure data storage, backup, and large-scale analysis. Furthermore, automated control mechanisms can be added to take immediate action during critical conditions. Overall, these enhancements will make the system more intelligent, scalable, and suitable for advanced naval and industrial applications.

APPENDICES

APPENDIX A SOURCE CODE

```
import numpy as np
import pandas as pd
import random
import io
from dash import Dash, html, dcc, Input, Output, State
import plotly.graph_objs as go
from collections import deque

# --- CONFIGURATION ---
VALID_USER = "lakshnaasree"
VALID_PASS = "031206"

# Theme Colors
SEA_BLUE_BG = "#001b2e"
CARD_BLUE = "#4382A6"
CYAN_GLOW = "#00e5ff"
ALERT_RED = "#d6527c"
OK_GREEN = "#00e676"

app = Dash(__name__)
app.title = "Mission Log"

# Data Buffers
history = {
    'temp': deque(maxlen=30), 'press': deque(maxlen=30),
    'volt': deque(maxlen=30), 'curr': deque(maxlen=30)
}
# Permanent log for CSV export
mission_log_data = []

# --- UI LAYOUT ---
app.layout = html.Div([

    # 1. LOGIN OVERLAY
    html.Div([
        html.Div([
            html.H2("DEEPSEA AUTHENTICATION", style={"color": CYAN_GLOW, "letterSpacing": "3px"}),
            html.P("ANALOG SCANNING MODE", style={"color": "#25aa2e", "fontSize": "12px"}),

            # USERNAME INPUT - Updated with blue text color
            dcc.Input(
                id="username",
                type="text",
                placeholder="Username",
                style={
                    "margin": "10px",
                    "padding": "12px",
                    "width": "250px",
                    "color": CYAN_GLOW, # Changes text color to Blue/Cyan
                    "fontWeight": "bold",
                    "border": "1px solid {CYAN_GLOW}",
                    "borderRadius": "5px",
```

```

        "backgroundColor": "#eff1f2" # Keeping white bg for contrast
    }
),

html.Br(),
dcc.Input(
    id="password",
    type="password",
    placeholder="Password",
    style={
        "margin": "10px",
        "padding": "12px",
        "width": "250px",
        "border": "1px solid #ccc",
        "borderRadius": "5px"
    }
),

html.Br(),
html.Button("INITIALIZE DIGITAL LOG", id="login-btn", n_clicks=0,
    style={"backgroundColor": CYAN_GLOW, "border": "none", "padding": "12px 30px",
"fontWeight": "bold", "marginTop": "20px", "cursor": "pointer"})
    ], style={"textAlign": "center", "padding": "60px", "backgroundColor": CARD_BLUE, "borderRadius":
"20px", "border": "1px solid {CYAN_GLOW}"})
    ], id="login-screen", style={"position": "fixed", "top": 0, "left": 0, "width": "100%", "height": "100%",
"backgroundColor": SEA_BLUE_BG, "display": "flex", "alignItems": "center", "justifyContent": "center",
"zIndex": "1000"}),

# 2. MAIN DASHBOARD
html.Div([
    html.Div([
        html.H1(" 🌊 DEEPSEA AI.CORE - MISSION CONTROL", style={"color": CYAN_GLOW,
"margin": "0"}),
        html.P("OPERATOR: LAKSHNAASREE | DIGITAL LOG ACTIVE", style={"color": "#00b4d8",
"fontSize": "12px"}),
        # DOWNLOAD BUTTON
        html.Button(" 📄 DOWNLOAD MISSION LOG (CSV)", id="btn-download", style={
            "marginTop": "10px", "backgroundColor": OK_GREEN, "border": "none", "padding": "8px 15px",
"borderRadius": "5px", "cursor": "pointer", "fontWeight": "bold"
        }),
        dcc.Download(id="download-log-csv"),
    ], style={"textAlign": "center", "padding": "20px"}),

    # Solution/Instruction Panel
    html.Div(id="solution-panel", style={
        "backgroundColor": "#002842", "margin": "0 40px 20px 40px", "padding": "15px",
        "borderRadius": "10px", "borderLeft": "5px solid {OK_GREEN}", "textAlign": "center"
    }),

    # Digital Log Numbers Grid
    html.Div([
        html.Div([html.Label("TEMP"), html.H2(id="d-temp")], className="log-card"),
        html.Div([html.Label("PRESS"), html.H2(id="d-press")], className="log-card"),
        html.Div([html.Label("VOLTS"), html.H2(id="d-volt")], className="log-card"),
        html.Div([html.Label("AMPS"), html.H2(id="d-curr")], className="log-card"),
    ], style={"display": "flex", "justifyContent": "center", "gap": "20px", "marginBottom": "20px"}),

    # Graphs Grid

```

```

html.Div([
    html.Div([dcc.Graph(id="g-temp", config={'displayModeBar': False})], className="graph-container"),
    html.Div([dcc.Graph(id="g-press", config={'displayModeBar': False})], className="graph-container"),
    html.Div([dcc.Graph(id="g-volt", config={'displayModeBar': False})], className="graph-container"),
    html.Div([dcc.Graph(id="g-curr", config={'displayModeBar': False})], className="graph-container"),
], style={"display": "flex", "flexWrap": "wrap", "justifyContent": "center", "gap": "15px"}),

    dcc.Interval(id="timer", interval=3000, n_intervals=0),
    dcc.Store(id='voice-store')
], id="main-content", style={"display": "none"}),

html.Div(id="voice-trigger", style={"display": "none"})

], style={"backgroundColor": SEA_BLUE_BG, "minHeight": "100vh", "color": "white", "fontFamily":
"monospace"})

# --- LOGIC & DIAGNOSTICS ---

def get_diagnostics(t, p, v, c):
    msg = ""
    instruction = "SYSTEMS NOMINAL: Maintain cruising depth."
    color = OK_GREEN
    problem = "None"

    if t > 32:
        problem = "Overheating"
        msg = f"Temperature high at {t} degrees. Initiate emergency cooling."
        instruction = "🔴 SOLUTION: Activate Primary Coolant Pump and Reduce Engine RPM."
        color = ALERT_RED
    elif p > 1.1:
        problem = "High Pressure"
        msg = f"Hull pressure critical at {p} atmospheres."
        instruction = "🔴 SOLUTION: Initiate Ballast Discharge and ascend to shallow waters."
        color = ALERT_RED
    elif v < 22:
        problem = "Low Voltage"
        msg = f"Voltage drop detected at {v} volts."
        instruction = "🔴 SOLUTION: Switch to Auxiliary Battery and disable non-essential systems."
        color = ALERT_RED
    elif c > 8:
        problem = "Power Surge"
        msg = "Current overload detected."
        instruction = "🔴 SOLUTION: Reset Main Circuit Breaker and check for short circuits."
        color = ALERT_RED

    return msg, instruction, color, problem

@app.callback(
    [Output("login-screen", "style"), Output("main-content", "style")],
    [Input("login-btn", "n_clicks")], [State("username", "value"), State("password", "value")]
)
def login(n, u, p):
    if n > 0 and u == VALID_USER and p == VALID_PASS:
        return {"display": "none"}, {"display": "block"}
    # Return initial style if login fails or hasn't happened
    return {"position": "fixed", "top": 0, "left": 0, "width": "100%", "height": "100%", "backgroundColor":
SEA_BLUE_BG, "display": "flex", "alignItems": "center", "justifyContent": "center", "zIndex": "1000"},
{"display": "none"}

```

```

@app.callback(
    [Output("g-temp", "figure"), Output("g-press", "figure"), Output("g-volt", "figure"), Output("g-curr",
"figure"),
    Output("d-temp", "children"), Output("d-press", "children"), Output("d-volt", "children"), Output("d-curr",
"children"),
    Output("voice-store", "data"), Output("solution-panel", "children"), Output("solution-panel", "style")],
    [Input("timer", "n_intervals")]
)
def update_all(n):
    # Simulate values
    t, p, v, c = round(random.uniform(20, 35), 1), round(random.uniform(0.9, 1.2), 2),
round(random.uniform(21, 26), 1), round(random.uniform(4, 9), 1)
    for key, val in zip(['temp', 'press', 'volt', 'curr'], [t, p, v, c]): history[key].append(val)

    voice, instr, col, prob = get_diagnostics(t, p, v, c)

    # Store in CSV log memory
    mission_log_data.append({
        "Timestamp": pd.Timestamp.now(),
        "Temp": t, "Pressure": p, "Voltage": v, "Current": c,
        "Status": prob, "AI_Instruction": instr
    })

    def make_fig(data, color):
        f = go.Figure(go.Scatter(y=list(data), mode='lines', line=dict(color=color, width=3), fill='tozeroy'))
        f.update_layout(paper_bgcolor='rgba(0,0,0,0)', plot_bgcolor='rgba(0,0,0,0)',
margin=dict(l=0,r=0,t=0,b=0),
                        xaxis=dict(visible=False), yaxis=dict(gridcolor="#004b73", tickfont=dict(color="white")))
        return f

    panel_style = {"backgroundColor": "#002842", "margin": "0 40px 20px 40px", "padding": "15px",
"borderRadius": "10px", "borderLeft": f"5px solid {col}", "textAlign": "center", "color": col}

    return make_fig(history['temp'], CYAN_GLOW), make_fig(history['press'], "#00b4d8"), \
        make_fig(history['volt'], "#ff003c"), make_fig(history['curr'], OK_GREEN), \
        f"{t} °C", f"{p} atm", f"{v} V", f"{c} A", voice, instr, panel_style

# CALLBACK FOR CSV DOWNLOAD
@app.callback(
    Output("download-log-csv", "data"),
    Input("btn-download", "n_clicks"),
    prevent_initial_call=True,
)
def download_log(n_clicks):
    df = pd.DataFrame(mission_log_data)
    return dcc.send_data_frame(df.to_csv, "mission_log_lakshnaasree.csv", index=False)

app.clientside_callback(
    "function(m){if(m){window.speechSynthesis.cancel();let s=new
SpeechSynthesisUtterance(m);s.rate=0.9;window.speechSynthesis.speak(s);}return ''}",
    Output("voice-trigger", "children"), Input("voice-store", "data")
)

if __name__ == "__main__":
    app.index_string = """
<!DOCTYPE html>
<html>

```

<head>

```
        "backgroundColor": "#eff1f2" # Keeping white bg for contrast
    }
),

html.Br(),
dcc.Input(
    id="password",
    type="password",
    placeholder="Password",
    style={
        "margin": "10px",
        "padding": "12px",
        "width": "250px",
        "border": "1px solid #ccc",
        "borderRadius": "5px"
    }
),

html.Br(),
html.Button("INITIALIZE DIGITAL LOG", id="login-btn", n_clicks=0,
    style={"backgroundColor": CYAN_GLOW, "border": "none", "padding": "12px 30px",
"fontWeight": "bold", "marginTop": "20px", "cursor": "pointer"})
    ], style={"textAlign": "center", "padding": "60px", "backgroundColor": CARD_BLUE, "borderRadius":
"20px", "border": "1px solid {CYAN_GLOW}"})
    ], id="login-screen", style={"position": "fixed", "top": 0, "left": 0, "width": "100%", "height": "100%",
"backgroundColor": SEA_BLUE_BG, "display": "flex", "alignItems": "center", "justifyContent": "center",
"zIndex": "1000"}),

# 2. MAIN DASHBOARD
html.Div([
    html.Div([
        html.H1("🌊 DEEPSEA AI.CORE - MISSION CONTROL", style={"color": CYAN_GLOW,
"margin": "0"}),
        html.P("OPERATOR: LAKSHNAASREE | DIGITAL LOG ACTIVE", style={"color": "#00b4d8",
"fontSize": "12px"}),
        # DOWNLOAD BUTTON
        html.Button("📄 DOWNLOAD MISSION LOG (CSV)", id="btn-download", style={
            "marginTop": "10px", "backgroundColor": OK_GREEN, "border": "none", "padding": "8px 15px",
"borderRadius": "5px", "cursor": "pointer", "fontWeight": "bold"
        }),
        dcc.Download(id="download-log-csv"),
    ], style={"textAlign": "center", "padding": "20px"}),

    # Solution/Instruction Panel
    html.Div(id="solution-panel", style={
        "backgroundColor": "#002842", "margin": "0 40px 20px 40px", "padding": "15px",
        "borderRadius": "10px", "borderLeft": "5px solid {OK_GREEN}", "textAlign": "center"
    }
]),
```

APPENDIX B – Screenshots

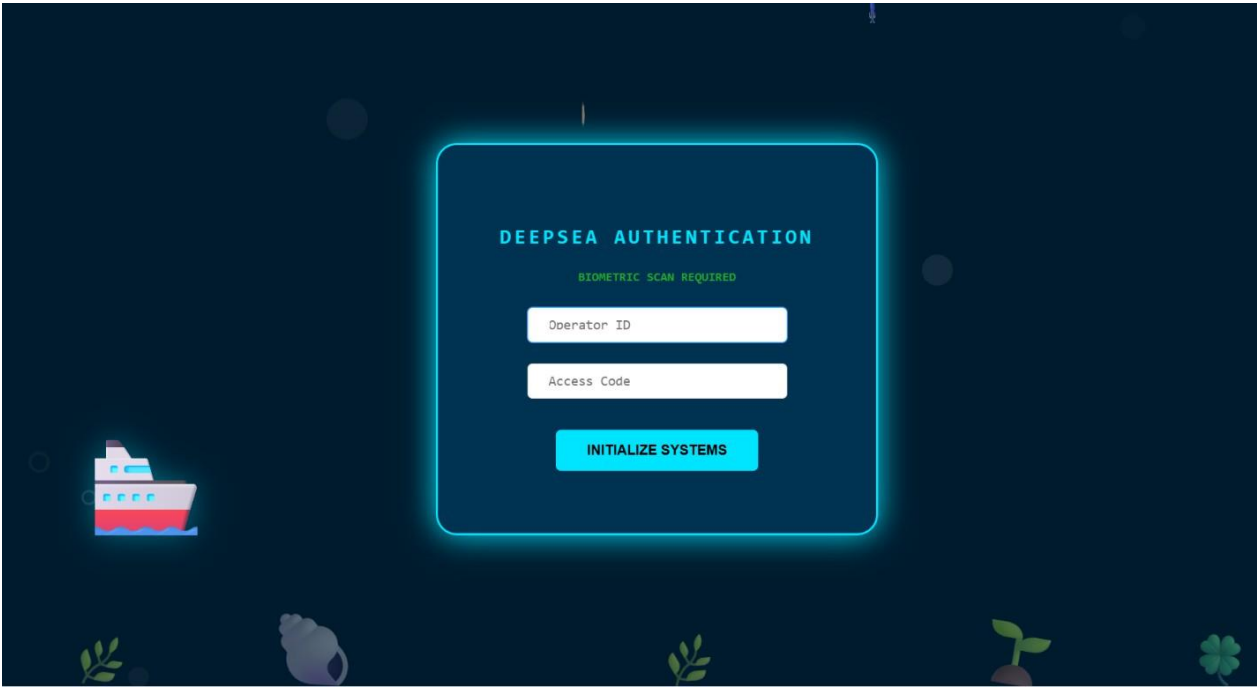


Figure B.1 Login Interface

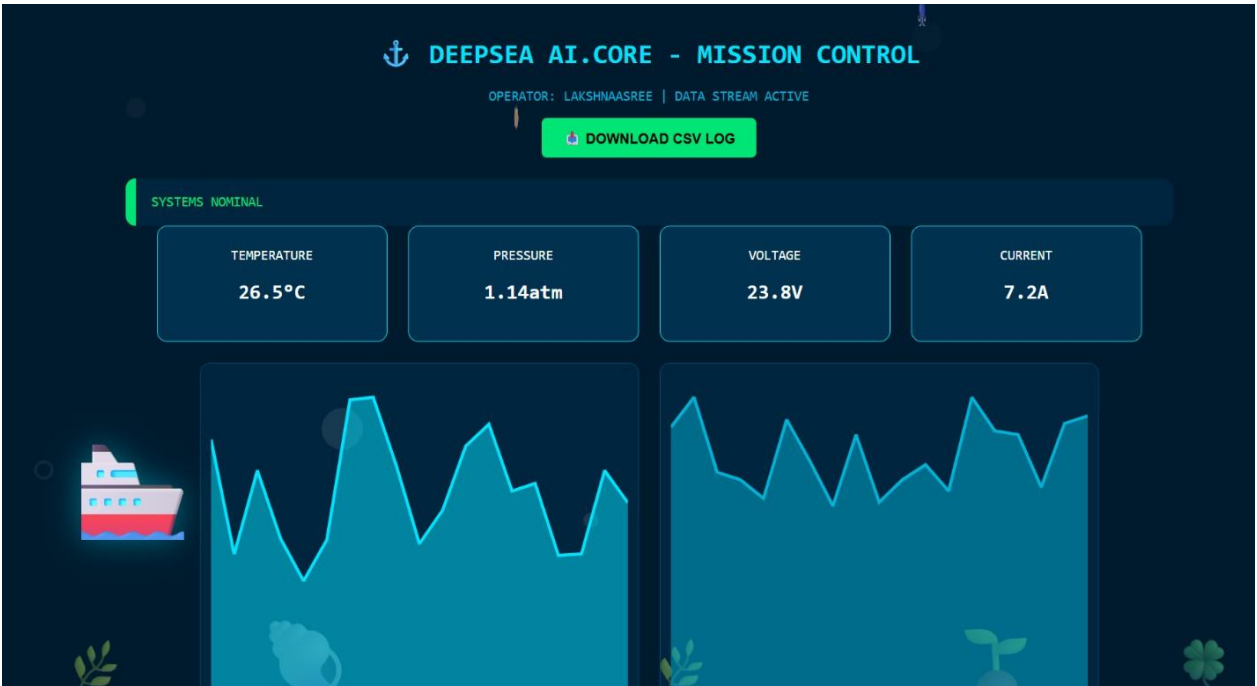


Fig B.2: Voice Alert Screen

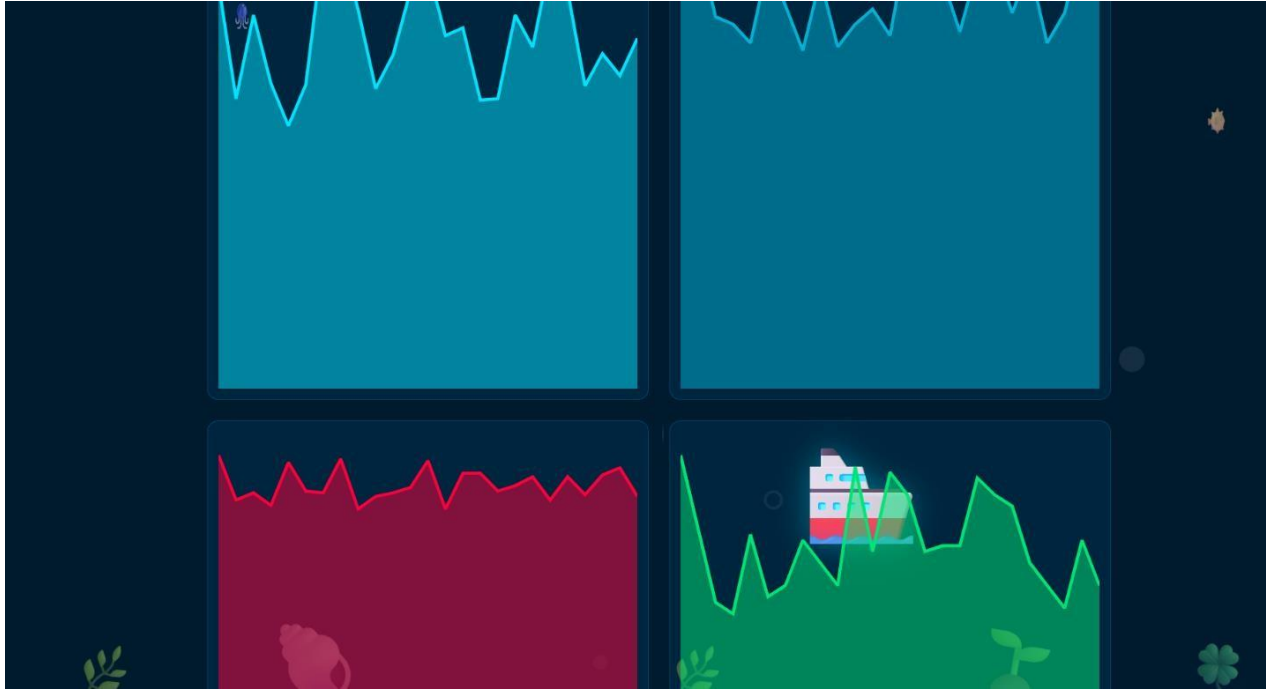


Figure B.3 Parameter Screen

	A	B	C	D	E	F	G	H	I	J	K
1	Timestamp	Temperature	Pressure	Voltage_V	Current_A	Status	Diagnostic_Instruction				
2	15-04-2026 22:45	31.4	1.11	25	7.9	Stable	SYSTEMS NOMINAL				
3	15-04-2026 22:45	22.5	1.19	22.5	6.6	Pressure	øÿ CAUTION: ASCEND NOW				
4	15-04-2026 22:45	29	0.99	22.9	5.3	Stable	SYSTEMS NOMINAL				
5	15-04-2026 22:45	23.7	0.97	22.2	5.1	Stable	SYSTEMS NOMINAL				
6	15-04-2026 22:45	20.4	0.92	24.6	6.5	Stable	SYSTEMS NOMINAL				
7	15-04-2026 22:45	23.6	1.13	23	5.4	Stable	SYSTEMS NOMINAL				
8	15-04-2026 22:45	34.5	1.02	22.9	5.6	Overheat	EMERGENCY: COOLING ON				
9	15-04-2026 22:45	34.7	0.9	24.8	6.4	Overheat	EMERGENCY: COOLING ON				
10	15-04-2026 22:45	29.4	1.09	22	6	Stable	SYSTEMS NOMINAL				
11	15-04-2026 22:45	23.3	0.91	22.7	5.6	Stable	SYSTEMS NOMINAL				
12	15-04-2026 22:45	25.9	0.97	22.9	7.7	Stable	SYSTEMS NOMINAL				
13	15-04-2026 22:45	30.9	1.01	23.2	6.2	Stable	SYSTEMS NOMINAL				
14	15-04-2026 22:45	32.6	0.94	24.7	7.6	Overheat	EMERGENCY: COOLING ON				
15	15-04-2026 22:46	27.4	1.19	22	7.2	Pressure	CAUTION: ASCEND NOW				
16	15-04-2026 22:46	28	1.1	24	6.2	Stable	SYSTEMS NOMINAL				
17	15-04-2026 22:46	22.4	1.09	24	6.3	Stable	SYSTEMS NOMINAL				
18	15-04-2026 22:46	22.5	0.95	23	6.3	Stable	SYSTEMS NOMINAL				
19	15-04-2026 22:46	29	1.12	23.3	7.5	Stable	SYSTEMS NOMINAL				
20	15-04-2026 22:46	26.5	1.14	23.8	7.2	Stable	SYSTEMS NOMINAL				
21	15-04-2026 22:46	33.1	1	22.5	7	Overheat	øÿ" EMERGENCY: COOLING ON				
22	15-04-2026 22:46	30.9	1.12	23.8	6	Stable	SYSTEMS NOMINAL				

Figure B.4 Status and Solution(Excel fo

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